$\boxed{\text{Case 4:}}$ The final digit of n is 4 or 6. Then the final decimal digit of n^2 is the final decimal digit of $4^2=16$ or $6^2=36$; namely, 6.

$\boxed{\text{Case 5:}}$ The final decimal digit of n is 5. Then the final decimal digit of n^2 is the final decimal digit of $5^2=25$, namely, 5.

$\boxed{\text{Case 6:}}$ The final decimal digit of n is 0. Then the final decimal digit of n^2 is the final decimal digit of $0^2=0$, namely 0.

Because we have considered all six cases, we can conclude that the final decimal digit of n^2, where n is an integer, is either 0, 1, 2, 4, 5, 6 or 9 □

$\boxed{\text{Example 6:}}$ Show that there are no solutions in integers x and y of $x^2+3y^2=8$. (Beautiful proof)

$\boxed{\text{Solution:}}$ We can quickly reduce a proof to checking just a few simple cases because $x^2>8$ when $|x|\geq 3$ and $3y^2>8$ when $|y|\geq 2$. This leaves the cases when x equals $-2, -1, 0, 1$ or 2 and y equals $-1, 0$ or 1. We can finish using an exhaustive proof. To dispense with the remaining cases, we note that possible values for x^2 are 0, 1, and 4, and possible values for $3y^2$ are 0 and 3, and the largest sum of possible values for x^2 and $3y^2$ is 7. Consequently, it is impossible for $x^2+3y^2=8$ to hold when x and y are integers. □

$\boxed{\text{Without Loss of Generality:}}$ In the proof in Example 4, we dismissed case (iii), where $x<0$ and $y\geq 0$, because it is the same as case (ii), where $x\geq 0$ and $y<0$, with the roles of x and y reversed. To shorten the proof, we could have proved cases (ii) and (iii) together by assuming, $\boxed{\text{without loss of generality}}$, that $x\geq 0$ and $y<0$. Implicit in this statement is that we can complete the case with $x<0$ and $y\geq 0$ using the same argument as we used for the case with $x\geq 0$ and $y<0$, but with the obvious changes.